



Figure 10-14 The carpus is placed through a range of motion. With the limb placed in a walking position, dorsal pressure is placed on the bottom of the foot to test for carpal hyperextension syndrome.



Figure 10-15 The radius and ulna are palpated. If pain is elicited in the midshaft of the radius in a skeletally immature large-breed dog, panosteitis is suspected.

osteosarcoma. Lameness and pain on palpation of the diaphyseal region of the radius or ulna in a young large-breed dog may indicate panosteitis ([Figure 10-15](#)).

The elbow is a difficult joint to assess because three bones are articulating together, and many soft-tissue structures originate in the region of the elbow. It is helpful to simultaneously palpate both elbows to detect subtle differences between them, such as mild joint effusion ([Figure 10-16](#)). The range of motion in flexion and extension is determined, and the joint is evaluated for crepitus. Any loss



Figure 10-16 Simultaneous palpation of both elbows allows for the detection of subtle abnormalities, such as mild joint effusion. The placements of hands and fingers are essentially mirror images of each other on the right and left elbows. Effusion of the caudolateral compartment in a 6-month-old large-breed dog may indicate the presence of an ununited anconeal process, whereas effusion of the craniomedial joint compartment may indicate a fragmented medial coronoid process or osteochondritis dissecans of the medial condyle of the humerus.

of motion, especially flexion, almost always indicates the presence of osteoarthritis. The medial and lateral humeral epicondyles and heads of the radius and ulna are palpated. The collateral ligaments of the elbow joint are evaluated with the elbow fully extended. The distal extremity is grasped and varus and valgus stresses are applied to evaluate the collateral ligaments. The elbow and carpus are then flexed to 90 degrees and the distal limb is internally and externally rotated. With this maneuver, it is possible to internally rotate the distal extremity and apply digital pressure on the medial portion of the joint. In many dogs with medial coronoid problems, pain may be elicited. Swelling in the caudolateral aspect of the elbow may indicate joint effusion associated with an ununited anconeal process, while swelling on the medial aspect of the elbow joint may indicate osteochondritis dissecans of the medial humeral condyle or a fragmented medial coronoid process of the ulna. With digital pressure, the effusion may be displaced, but the effusion reappears after the digital pressure is released.

The humerus is palpated for swelling and pain that may result from panosteitis or neoplasia. Caution should be exercised in interpreting pain when palpating the region of the radial nerve ([Figure 10-17](#)). Direct pressure on the



Figure 10-17 Caution should be exercised when palpating the distal humerus to be certain that compression of the radial nerve by the examiner is not causing pain and discomfort. The general area of the radial nerve is indicated by the finger.

nerve may elicit a painful response unrelated to any pathologic finding of the humerus. Anatomically, an imaginary line drawn along the dorsal border of the antebrachium and continued to the humerus with the elbow held in mild flexion will approximate the location of the radial nerve as it crosses over the humerus.

The scapulohumeral or shoulder joint should be flexed and extended. The relationship of the acromial process to the greater tubercle of the humerus should be palpated. An abnormality of this relationship may indicate a shoulder luxation, either medial or lateral. In the absence of trauma, pain on extension of the shoulder joint of a young dog is almost pathognomonic for osteochondritis dissecans of the caudal aspect of the humeral head (**Figure 10-18**). Similarly, pain may be elicited in a dog with osteochondritis dissecans if the shoulder joint is simultaneously flexed and internally rotated. Pain may be elicited with simultaneous flexion of the shoulder joint and extension of the elbow if biceps tenosynovitis is present (**Figure 10-19**). This maneuver places increased stress on the biceps tendon as it traverses the bicipital groove in the proximal humerus. Direct palpation of the biceps tendon in this region, located cranial and medial to the greater tubercle, may also elicit pain.

In addition, varus and valgus stresses should be applied to the shoulder to check the collateral support. Increasingly, medial glenohumeral ligament instability is being recognized. The examiner uses the thumb and forefinger of the same hand to grasp the acromion process and place



Figure 10-18 Pain caused by extension of the shoulder joint in a skeletally immature large-breed dog is consistent with osteochondritis dissecans of the caudal head of the humerus.



Figure 10-19 Pain caused by shoulder flexion and simultaneous extension of the elbow is likely due to biceps tenosynovitis. Direct palpation of the region of the biceps tendon just medial to the greater tubercle of the proximal humerus may also be painful.

medial pressure on it to prevent movement of the scapula.¹ The examiner then grasps the limb at the level of the distal humerus using the opposite hand. With the elbow and shoulder held in extension, the shoulder joint is abducted to its physiologic limit (**Figure 10-20**). Normal dogs have an abduction angle of 30-35 degrees, whereas dogs with medial glenohumeral instability have increased abduction



Figure 10-20 Shoulder abduction is performed to detect the presence of medial glenohumeral instability. Note that the shoulder and elbow joints are held in extension while the shoulder is abducted.



Figure 10-21 Palpation of the scapula is performed to evaluate for fractures or, rarely, luxation of the scapula.

and usually pain at the end range of abduction. Dogs with chronic lameness of a forelimb, unrelated to medial glenohumeral instability, with muscle atrophy may have increased shoulder abduction, but do not usually have pain at the end range of abduction. In some dogs with chronic scapulohumeral instability, it is possible to elicit a drawer motion of the humerus in relation to the scapula.

The scapula should be palpated for fractures or pain, especially the spine of the scapula (Figure 10-21). The body of the scapula should be moved in a dorsal direction to determine if a scapular luxation exists, although this is a rare condition.

Other Tips

In general, the best systematic approach to the orthopedic examination is to examine the normal limbs first and the affected limb last. In many cases subtle problems in other limbs may be masked or not detected if attention is initially placed on the most obvious problem. Also, many animals become tense throughout the body if they experience discomfort as a result of palpating a painful area on the affected limb, and a thorough, accurate examination may no longer be possible. Tranquilization and sedation should be avoided until after the initial examination has been completed so that discomfort during palpation is not masked. Similarly, it is best to avoid antiinflammatory drugs for 24 to 48 hours before examination. Diagnostic joint blocks should also be considered. These can be helpful in prioritizing problems if more than one joint is involved, such as in osteoarthritis of the tarsocrural joint and chronic rupture of a cranial cruciate ligament. Tranquilization may facilitate examination of large, athletic dogs in which muscle tone or an anxious patient results in difficulty in assessing certain aspects of the examination, such as an Ortolani sign or cranial drawer motion.

Arthrocentesis

In some cases, arthrocentesis of one or more joints may be performed by a veterinarian. The joint should be identified and will typically have some effusion or an increased quantity of synovial fluid. The area should be clipped and prepared using aseptic technique. A 20- or 22-gauge needle and 3-mL syringe are used. The affected joint should be flexed and extended several times before arthrocentesis. This helps to dislodge any bacteria from the synovium if an infection is suspected. The needle is inserted without the syringe attached, and the veterinarian waits for synovial fluid to drop into a small EDTA tube. The needle should be positioned downward to take advantage of gravity to encourage the flow of synovial fluid. This avoids direct aspiration of the joint and iatrogenic hemorrhage. If no synovial fluid drips from the needle, then the syringe is attached and gentle aspiration is attempted. The fluid should be cultured if sepsis is suspected. Two samples are obtained, one for immediate culture, and the other for culture in blood culture media for 24 hours. This will allow the recovery of relatively low numbers of bacteria to be detected. In many cases, only several drops of fluid may be obtained, precluding a full cell count and evaluation. However, cytology may be performed to determine relative cellularity and the distribution of cell types. Synovial fluid analysis is most useful to differentiate inflammatory and noninflammatory joint disease. Table 10-1 indicates the most common general joint conditions and the characteristics of synovial fluid analysis.

Table 10-1 Synovial Fluid Changes in Canine Arthritis

Condition	Nucleated Cells/mm ³	Differential Mononuclear Cells	Differential Neutrophils
Normal	250-3000	94-100	0-6
DJD	1000-5000	88-100	0-12
Erosive (rheumatoid-like)	8000-38,000	20-80	20-80
Nonerosive	4400-371,000	5-85	15-95
Septic	40,000-267,000	1-10	90-99

DJD, Degenerative joint disease.

From Piermattei DL, Flo GL: Brinker, Piermattei, and Flo's handbook of small animal orthopedics and fracture repair, ed 3, Philadelphia, 1997, WB Saunders.

Neurologic Examination

Dogs suspected of having a neurologic condition should have a complete neurologic evaluation. In addition, during evaluation of small animals for lameness or other musculoskeletal abnormalities, the neurologic system must always be examined because some “lamenesses” are actually neurologic conditions. A complete neurologic examination should be performed to help determine the contribution of the nervous system to the condition being evaluated, and to help identify other problems that may be occurring concurrently. In addition, a neurologic evaluation should be performed in trauma patients, and owners should be made aware of conditions before undertaking any treatment of traumatic musculoskeletal conditions. A neurologic examination should be performed in a systematic, sequential manner. By always performing a neurologic evaluation in a consistent fashion, the examiner can ensure that all aspects of the nervous system are evaluated and that nothing is omitted. The neurologic examination should always be performed in a calm, quiet place with minimal distractions. It is important that the patient not be sedated or tranquilized before the examination to help minimize erroneous interpretations of pain sensation or reflexes.

General Neurologic Examination

The neurologic examination begins when an animal is first presented to the hospital. The general appearance of the patient may initially be evaluated at the time the history is taken, even before a physical examination is performed. Behavior, posture, and voluntary movements should be noted as the animal is allowed to freely move around the examination room. During this time, the interactions between the patient and owner should be observed, as this can give valuable information about the patient's mentation. An animal's behavior may be normal, aggressive, excited, apathetic, or depressed. A change in mentation should be noted because this may indicate abnormal brain function or other problems. Posture and voluntary movement may also indicate abnormal forebrain or brainstem

function, but may also indicate a spinal cord lesion or other neurologic conditions. For an animal to have normal posture, normal peripheral sensory inputs must be processed in the higher centers of the brain including the cerebrum, the cerebellum, and the brainstem. Conditions such as head tilts, tremors, or falling to one side may indicate a problem in any one of these locations.

Nonambulatory animals may be paretic or paralyzed. Either condition causes obvious and markedly abnormal posture in patients. Paretic animals cannot walk without assistance, and although patients can still voluntarily move their legs, limb motion is weak. A lesion causing paresis is considered less severe than one causing paralysis in which voluntary movement is no longer possible. Although paresis may be caused by many conditions, it is often a sign of a less severe lesion than would be seen in a paralyzed animal. If a paretic lesion worsens, the condition often progresses to complete paralysis. The importance in differentiating paresis from paralysis comes into play when determining prognosis. When examining quadrupeds with spinal lesions, the examiner should keep in mind that paresis and paralysis do not always affect all four limbs equally. When an animal is unable to use or move its limbs, it is said to be *paralyzed*. When an animal is unable to walk because of paresis in all four limbs, the term *tetraparesis* is used. *Hemiparesis* and *paraparesis* are terms used to describe paresis on one side of the body or of the hindlimbs, respectively, while other limbs have normal function. These presentations are typically seen when a lesion affects nerves to a particular limb without affecting the innervation of other limbs, or when a lesion affects only one side of the spinal cord. A typical example is a thoracolumbar spinal fracture leading to paraparesis or paralysis of both hindlimbs while forelimb function remains unaffected.

After the history is recorded and the general appearance of the animal is evaluated, the patient's gait should be evaluated if the patient is capable of ambulating and if trauma to the vertebral column is not present. A harness should be used for any patient in which head or neck lesions are suspected. In preparation for this aspect of the evaluation, a handler is instructed to gait the animal at a

walk and a trot. Although an ataxic or circling animal may not need to be trotted, other conditions such as hypometria or hypermetria may not be evident until the speed of the gait is increased. Gait evaluation should always be performed on a surface with adequate traction and should involve evaluation while the patient ambulates toward and away from the examiner. Several passes at a walk and trot should be performed to detect any minor deficits that may be present. Special attention should focus on each limb and not just the limb or limbs in question to help ensure that all four limbs are properly examined. Some patients that have severe ataxia or a tendency to circle can be difficult to walk in a straight line. It may be beneficial in these patients to allow them to ambulate freely around a large confined space to more adequately characterize their gait.

After behavior, posture, voluntary movements, and gait have been assessed, the physical examination is initiated. Palpation allows the examiner to inspect muscles, bones, and other soft tissues for subtle abnormalities. Palpation is performed in a systematic way beginning at the head, moving down the neck, and proceeding to the trunk and extremities. Again, by following the same procedure for each patient, the examiner minimizes omissions and is able to perform a consistent evaluation. One side of the body is compared with the other to evaluate symmetry. In patients with one side or limb more severely affected, it may be beneficial to palpate and examine the more normal side or limb first. Careful attention should be paid to muscle size, tone, and strength. Deviation of joints or limbs, abnormal body carriage, or worn toenails could indicate a neurologic problem. Because intervertebral disk herniation and other forms of spinal cord trauma may be painful, the presence of pain during physical examination not only may help identify possible differential diagnoses for a condition, but also may help to localize the lesion.

Spinal Reflexes

Spinal reflexes are used to test the sensory and motor components of a reflex arc. In general, reflexes may be (1) decreased or absent (hyporeflexia), indicating either partial or complete loss of sensory or motor nerve function to a particular region of the body (lower motor neuron [LMN] disease); (2) normal, indicating that no neurologic abnormalities exist within the reflex arc; or (3) increased or exaggerated (hyperreflexia), indicating that the inhibitory neurons from the brain and spinal cord cranial to the LMNs have been affected (upper motor neuron [UMN] disease). In general, spinal reflexes are graded using the following scheme: absent (0); decreased (+1); normal (+2); exaggerated (+3); or very exaggerated or clonus (+4).

Myotatic Reflexes

Myotatic reflexes are stretch reflexes that cause muscle contraction after the muscle is stretched. These are local reflex arcs that are used for body posture and movement

and do not rely on cerebral input for function. The reflex arc consists of a sensory neuron that responds to the stretch of a muscle, and a motor neuron that in turn causes muscle contraction. In general, if only one limb is affected, a peripheral nerve injury is suspected, whereas a spinal cord lesion usually produces bilateral deficits. The most reliable myotatic reflex is the patellar reflex. This reflex is performed with a relaxed patient in lateral recumbency and the pelvic limb held in a slightly flexed position. With the knee supported, the patellar ligament is briskly tapped with a pleximeter (Figure 10-22). As long as the leg is not already in full extension, suddenly stretching the patellar ligament evokes a reflex arc that causes sudden extension of the stifle. Typically, if this reflex is diminished or absent, an LMN lesion is localized to spinal cord segments L4 to L6 (femoral nerve) or to peripheral sensory or motor nerves. If hyperreflexia is present, a UMN lesion involving the spinal cord cranial to L4 is suspected.

The cranial tibial reflex is tested with the animal in the same position as for the patellar reflex. The pleximeter is used to strike the muscle belly of the cranial tibial muscle, eliciting flexion of the hock. This reflex is localized to the L6 to L7 segments of the spinal cord (peroneal branch of the sciatic nerve). The gastrocnemius reflex can then be tested while the animal is still in lateral recumbency. The hock can be slightly flexed, and either the gastrocnemius tendon can be struck or the muscle belly can be struck eliciting extension of the hock. This reflex involves the



Figure 10-22 A patellar reflex is elicited with the patient relaxed and in lateral recumbency by tapping the patellar ligament with a pleximeter.

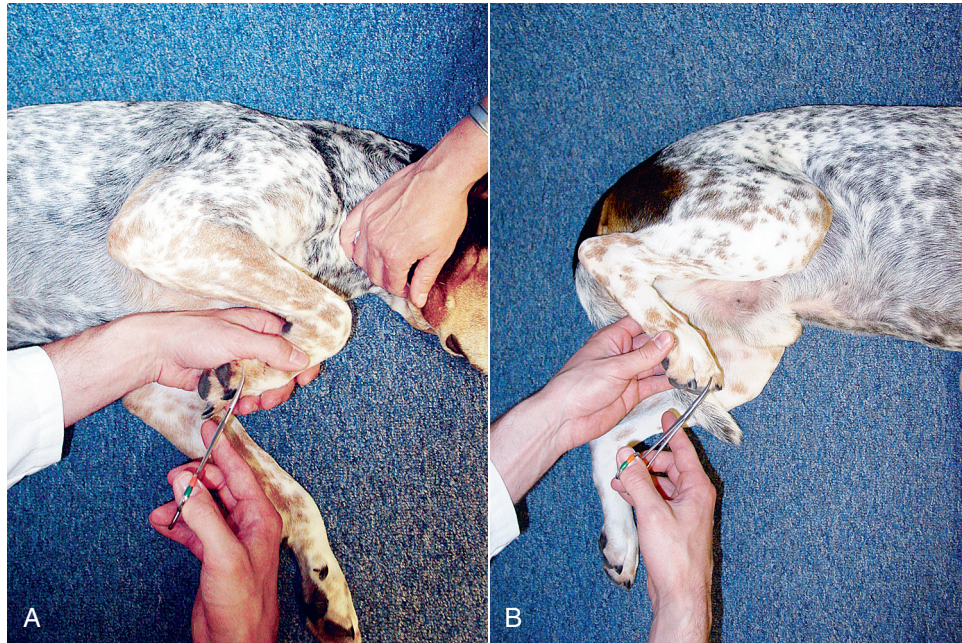


Figure 10-23 Eliciting a flexor reflex of a forelimb (A) and a hindlimb (B). Note flexion of the shoulder, elbow, and carpus in the forelimb, and the hip, stifle, and hock in the rear limb.

L7-S1 segments of the spinal cord (tibial branch of the sciatic nerve). The cranial tibial and gastrocnemius reflex are much less reliable than the patellar reflex. If these reflexes are present, the reflex arc is intact. However, the inability to obtain a reflex does not necessarily indicate that there is no intact pathway.

In the thoracic limb, the reflexes tested include the extensor carpi radialis, triceps, and biceps reflex. For the extensor carpi radialis reflex, the limb is held with the elbow in slight flexion and the carpus slightly flexed as well. The muscle belly is struck, which should cause extension of the carpus. This reflex is localized to spinal cord segments C7-T1 (radial nerve). The triceps reflex also tests these segments. The limb is held in the same position and the triceps tendon is struck just proximal to its insertion, resulting in contraction of the muscle. Finally, the biceps reflex is tested by extending the elbow while palpating the biceps tendon just proximal to its insertion. The pleximeter is used to strike the examiner's finger, resulting in contraction of the biceps muscle. This reflex involves the segments from C6-C8 (musculocutaneous nerve). Myotatic reflexes in the thoracic limbs can be difficult to interpret, as many normal animals have decreased responses. The flexor reflex gives the most information about nerve function in this area.

Flexor Reflexes

Flexor reflexes are used to evaluate nerve function in both the thoracic and pelvic limbs. These reflexes are useful in the normal animal to help prevent injury from noxious

stimuli and to allow withdrawal of the limb away from a noxious stimulus. Briefly, the animal is positioned in lateral recumbency and the limb is positioned in a neutral position. The animal should be relaxed and comfortable. A noxious stimulus is applied to the foot, stimulating sensory neurons (Figure 10-23). Only the amount of stimulus required to elicit a response should be used. Initially, the skin on a toe or between toes is pinched. If no response is elicited with a mild stimulus, a hemostat placed across the nail bed of a toe is used before concluding that there is a negative response. Both medial and lateral aspects of the foot should be tested. The motor pathways of flexor reflexes are more complex than those of myotatic reflexes because motor neurons for all flexor muscles in the affected limb are activated and the entire limb is flexed. An example is a flexor reflex of the pelvic limb of a dog, which causes flexion of the hock, stifle, and hip.

Autonomous Zones

Because specific peripheral sensory nerves innervate various portions of the limbs, the peripheral nerves and their spinal cord segments may be evaluated to determine the level of injury. In the thoracic limb, the dorsal surface of the foot is innervated by the radial nerve (C7-T1), the medial surface of the antebrachium is innervated by the musculocutaneous nerve (C6-8), and the caudolateral surface of the foot is innervated by the ulnar nerve (C8-T2). In the pelvic limb, the most medial toe is innervated by the saphenous nerve, a branch of the femoral nerve (L4-L6), the plantar surface is innervated by the tibial

branch of the sciatic nerve (L7-S1), and the rest of the foot is innervated by the peroneal branch of the sciatic nerve (L6-L7). Variable responses to noxious stimuli in different regions of the limb help to localize the lesion to a particular region.

Crossed-Extensor Reflex

The crossed-extensor reflex is a normal reflex in a standing animal. The reflex causes extension of one limb when the contralateral limb is flexed. In a recumbent animal, this reflex is normally inhibited by descending inhibitory pathways. To elicit the reflex, the animal is placed in lateral recumbency, and the flexor reflex is evoked from the upper limb. Extension of the contralateral limb while the patient is in lateral recumbency signals a lesion in the inhibitory pathways or pathologic condition of the UMN.

Perineal Reflex

The perineal reflex is elicited by tactile stimulation of the perineal region. It is important to test both the left and right side, as some lesions may cause only unilateral signs. The normal response of this reflex consists of contraction of the anal sphincter (anal winking) and ventral flexion of the tail. The sensory and motor components of this reflex are innervated by the pudendal nerve; thus an absent or decreased response indicates a lesion of the pudendal nerve or sacral spinal cord segments S1-S3. Lesions in the sacral spinal cord or pudendal nerves also often result in decreased urethral sphincter tone and an LMN bladder. This reflex can also be noted when taking the temperature of the patient during the initial physical examination.

Cutaneous Trunci Reflex

The cutaneous trunci reflex (sometimes called the *panniculus reflex*) is a relatively complex reflex elicited by applying a noxious stimulus to the skin, which stimulates the superficial spinal nerves innervating a particular region or autonomous zone (dermatome) of the dermis and elicits a motor response, which is seen as a skin twitch. Typically, fingers or hemostats are used to pinch the skin and dermis just lateral to the dorsal midline, beginning in the area of the sacrum and moving cranially until a response is elicited. Where this response starts varies between patients, but usually is in the mid to caudal lumbar region. When the cutaneous trunci reflex is stimulated, a reflex arc is initiated, with the afferent nerves of that particular region of the skin stimulated and a signal transmitted to the spinal cord. The signal is transmitted cranially to the C8-T1 spinal cord segments, where the nerves synapse with the efferent nerves of the lateral thoracic nerve. The impulse causes contraction of the cutaneous trunci muscles to elicit a skin twitch. Spinal nerve roots to autonomous zones travel from cranial to caudal; therefore this reflex usually tests spinal cord segments or nerves approximately two vertebral bodies cranial to the noxious stimulus. Thus a

deficit in the cutaneous trunci response at a specific location indicates a spinal cord lesion approximately two vertebral bodies cranial to the site of the skin pinch. The cutaneous trunci reflex is most reliable in the middle of the back. Testing over the cervical, caudal lumbar, and sacral regions is far less reliable.

Sensation

The presence or absence of sensation in each appendage must be assessed. Unless peripheral nerve injury is suspected, pain sensation is usually absent only when voluntary motion is also absent. This typically occurs in the presence of severe spinal cord injury. In general, if an animal can move its limbs, sensation is usually present. In paralyzed animals the presence or absence of pain sensation is very important because absence of deep pain sensation may indicate severe and often irreversible spinal cord damage.

Deep pain sensation is assessed concurrently with the flexor reflexes. Animals with severe damage to the spinal cord cranial to the nerve cell bodies and peripheral nerves of the tested flexor reflex may still have an intact flexor reflex, but may not transmit the signal to the brain for the interpretation of pain. Intact pain perception is indicated by a cerebral response to the pain, such as vocalization, turning the head toward the stimulated limb, or an attempt to move away from the stimulus. In very stoic patients, the response may be seen as a cessation of panting or movement of the eyes. The absence of deep pain perception usually indicates severe spinal cord damage, but it does not indicate whether the lesion is UMN or LMN. The presence of a hyperreflexic or hyporeflexic flexor reflex indicates whether the lesion is UMN or LMN. It must be emphasized that a patient may still retain a positive flexor reflex, but have loss of deep pain sensation. This situation occurs relatively commonly in dogs with serious intervertebral disk extrusion in the thoracolumbar region. Therefore a conscious response must be obtained to determine if the pain pathway is intact.

Postural Reactions

Postural reactions are used by animals to maintain an upright position and normal body posture. Under normal circumstances, an animal knows the position of each limb in space and will appropriately shift its weight to avoid falling. Postural reactions involve spinal reflexes, as well as input from higher centers, such as the brain and spinal cord, to maintain proper body position. Several postural reactions should be tested during a complete neurologic examination. Although these tests involve different limbs, all test the integration of brain, spinal cord, and peripheral nerves. Abnormal postural reactions indicate a neurologic abnormality, but provide the examiner with limited information regarding the specific location of the lesion. Other portions of the neurologic examination will help to localize



Figure 10-24 Hopping performed to assess postural reactions in a forelimb.



Figure 10-25 Wheelbarrowing to assess proprioceptive function of the forelimbs.

the lesion. Common postural reaction tests are described in the following sections.

Hopping

Hopping may be performed on thoracic and pelvic limbs. The animal is supported, and weight is placed on one limb with the other limbs supported so they bear no weight (Figure 10-24). The animal is then moved laterally in a smooth, quick fashion. The animal should hop on the limb to prevent falling and collapse. Caution should be used in animals with muscle weakness because these animals might have a normal postural reaction to hopping, but may not have the muscle strength to support their weight. In these situations, the animal may be supported to allow hopping, even if it is weak. In larger dogs hopping may be tested by supporting a thoracic limb completely and hopping the animal in the thoracic region using the pelvic limbs as a pivot. This technique allows for adequate testing of the response, without the examiner having to fully support the patient's weight. This can be repeated in the pelvic limbs, using the thoracic limbs as the pivot point.

Wheelbarrowing

Wheelbarrowing is performed by lifting the caudal abdomen and pelvic limbs of the dog and allowing weight bearing on the forelimbs (Figure 10-25). In this position, a dog with normal neurologic function should be able to move forward as the dog is pushed forward. Muscle weakness may affect wheelbarrowing and postural reactions, requiring some support.



Figure 10-26 Hemiwalking.

Hemiwalking

The hemiwalking test is performed by lifting ipsilateral forelimbs and hindlimbs of the patient, forcing weight bearing on the other two contralateral limbs (Figure 10-26). The dog is moved laterally and should be able to walk on two legs without falling. Weak animals may require some support.

Conscious Proprioception

Conscious proprioception is used to test an animal's awareness of limb position in space without visual input. It is unnatural for a dog to walk or stand on the dorsum of the paw. To test conscious proprioception, a knuckling test is



Figure 10-27 Conscious proprioception is evaluated by knuckling over the dorsum of the paw. A neurologically normal patient will quickly right the paw to a proper position.

performed on each of the four limbs. With the animal supported, the paw is picked up and turned over so that the dorsum of the paw contacts the ground (Figure 10-27). Generally, normal animals immediately pick up the paw and place it into a normal position. Normally, a patient will not stand on the dorsum of the paw to support weight. If the paw is not properly replaced, a neurologic deficit is usually present. In general, animals with mild neurologic lesions may have proprioceptive deficits, but may not show other obvious abnormal signs; in some cases, motor activity may remain relatively normal until the lesion progresses.

Tactile Placing

Tactile placing tests are performed by covering the eyes, lifting the dog off of the ground, and bringing the forelimbs toward a table or stable surface. After sensing the surface, the normal response to a tactile placing test is to immediately place both forelimbs onto the surface in a weight-bearing position. A normal tactile placing response relies on normal sensory and motor pathways of the thoracic limbs. Also, normal cerebrocortical function is required to receive and process the information.

Visual placing tests are performed in the same manner as a tactile placing test except that the patient's eyes are not covered. During this test, it is normal for a suspended animal to reach for a surface even before touching it. Although this test still relies on a normal cerebral cortex and intact motor function to the thoracic limbs, sensory

input is via the visual pathways rather than the tactile sensation of the paw.

Cranial Nerve Examination

A brief cranial nerve examination should be performed during a neurologic evaluation. Animals possess 12 cranial nerves (Table 10-2). These nerves are mainly involved with functions of the head, and abnormal function may indicate a brain lesion. As with other neurologic tests, cranial nerve examination should be performed in a systematic manner and usually does not add more than a few minutes to the examination. Cranial nerve examination is briefly discussed; the nerves evaluated are indicated in parentheses after each portion of the test.

The patient's head is inspected for muscle atrophy and abnormal jaw tone (cranial nerves VII [facial nerve] and V [trigeminal nerve], respectively). While the eyes are covered, the skin around the ears, eyes, and nose is stimulated to elicit a twitch or other facial movement (cranial nerves V and VII). The lateral and medial canthus of each eye is stimulated to elicit a blink response. A menace response is elicited by quick hand movements toward each open eye. Caution must be used in this interpretation, as an air current created by the examiners hand has the potential to stimulate a response as well as if the examiner unintentionally touches the whiskers of the patient. In adult animals, anticipation of ocular injury should cause momentary closure of the affected eye (cranial nerves II [optic nerve] and VII). The menace response is a learned response and is generally not able to be elicited until approximately 10-12 weeks of age. A pupillary light response is tested by shining a bright light into each eye while monitoring for pupillary constriction (cranial nerves II and III [oculomotor nerve]). Direct and consensual pupillary light responses should be tested. To elicit a physiologic nystagmus response, the patient's head is moved from side to side. The fast phase of the nystagmus should be in the direction of the movement (cranial nerves III, IV [trochlear nerve], and VI [abducens nerve]). The mouth is then opened and the gag reflex is tested (cranial nerves IX [glossopharyngeal nerve] and X [vagus nerve]). At the same time, symmetry of the tongue should be observed, which also can occur when the patient licks (cranial nerve XII [hypoglossal nerve]). Palpation of the neck muscles for the presence of atrophy helps to evaluate cranial nerve XI (accessory nerve). The sense of smell may be tested by monitoring for a response to a food odor such as a favorite treat positioned near the nose (cranial nerve I [olfactory nerve]), although this reflex is rarely tested.

Interpreting the Neurologic Examination

After the neurologic examination is complete, the results of each individual test should be reviewed to help determine the presence or absence of a neurologic abnormality and to help the examiner to localize the lesion. In general,

Table 10-2 Cranial Nerves and Their Functions

CN I: Olfactory nerve	Sensory pathway for the sense of smell
CN II: Optic nerve	Sensory pathway for vision and pupillary light reflexes
CN III: Oculomotor nerve	Motor pathway to all extrinsic ocular muscles except the lateral rectus, retractor bulbi, and dorsal oblique muscles
	Motor pathway for pupillary constriction
CN IV: Trochlear nerve	Motor pathway to the dorsal oblique muscle of the eye
CN V: Trigeminal nerve	Motor pathway to masticatory muscles
	Sensory pathway to the mouth and face
CN VI: Abducent nerve	Motor pathway for the lateral rectus and dorsal oblique muscles of the eye
CN VII: Facial nerve	Motor pathway to the muscles of facial expression
	Sensory pathway for taste to the rostral two thirds of the tongue and the palate
CN VIII: Vestibulocochlear nerve	Vestibular portion—Proprioceptive role for equilibrium and posture
	Cochlear portion—Sensory role in hearing
CN IX: Glossopharyngeal nerve	Sensory and motor innervation for swallowing and the gag reflex
	Sensory pathway for taste to the caudal third of the tongue
CN X: Vagus nerve	Sensory and motor innervation for swallowing, the gag reflex, laryngeal function, and vocalization
CN XI: Spinal accessory nerve	Motor pathway to the muscles of the neck
CN XII: Hypoglossal nerve	Motor innervation to the muscles of the tongue

CN, Cranial nerve.

C1-C5	C6-T2	T3-L3	L4-S3
UMN Forelimbs UMN Rear Limbs	LMN Forelimbs UMN Rear Limbs	Normal Forelimbs UMN Rear Limbs	Normal Forelimbs LMN Rear Limbs
- If lesion is L4-L5, LMN signs to femoral nerve distribution - If lesion L6-S2, Normal femoral nerve function, LMN to sciatic nerve			

Figure 10-28 Appearance of upper motor or lower motor neuron signs associated with lesions of various spinal cord segments or their nerve roots.

animals with progressive neurologic disease initially have decreased proprioception, followed by loss of motor function, and, in severe cases, a patient may progress to loss of superficial and finally, deep pain. If patients recover, pain sensation generally returns first, followed by return of some motor function, and with near-complete recovery, proprioceptive function returns last.

Interpretation of the neurologic examination is important to localize the lesion so that additional diagnostic tests can be performed to evaluate a particular area. The hallmarks of LMN signs are hyporeflexia, weakness, and reduced resting tone of muscles, and a decreased ability to generate tone with limb movement. The indications of UMN signs are hyperreflexia, maintenance of strength, increased resting tone of muscles, and spasticity of the limbs.

Lesions of spinal cord segments or nerves of C1-C5 result in UMN signs to all four limbs (Figure 10-28). In

some cases, the clinical signs may be more obvious in the hindlimbs because the spinal tracts to these limbs are located more peripherally in the ventral spinal cord, where compression by a space-occupying mass, such as a protruding disk, may occur. Lesions of spinal cord segments or nerves of C6-T2 will cause LMN signs to the forelimbs and UMN signs to the hindlimbs. The forelimbs will be normal if the lesion is caudal to T2. Lesions between T3 and L3 will result in UMN signs to the hindlimbs. Lesions located from L4 to S3 will result in LMN signs to the hindlimbs.

Ancillary Tests

Thorough orthopedic and neurologic examinations help to isolate the area of interest to a smaller region. This ultimately saves time and expense and will likely result in a more accurate diagnosis. Ancillary tests to aid

in a definitive diagnosis include radiographs, magnetic resonance imaging (MRI), computed tomography (CT), myelography, electrodiagnostics, and nuclear medicine studies.

Radiographs are the most commonly used test. It is important to always take at least two views at 90-degree angles. Most commonly, lateral and cranial-caudal views are obtained of the region of interest. Comparison films of the opposite limb may be useful to evaluate subtle lesions. Occasionally, stress views of the tarsus or carpus may be helpful. Specialty clinics and university teaching hospitals may have the capability to perform advanced diagnostic

tests such as MRI or CT. These studies may give enhanced resolution and information, particularly of soft tissue structures, and of the three-dimensional nature of conditions. Nuclear studies may indicate regions of inflammation and metabolic activity in soft tissues or bone, but are available in only a few locations.

REFERENCE

1. Cook JL, Renfro DC, Tomlinson JL, Sorensen JE: Measurement of angles of abduction for diagnosis of shoulder instability in dogs using goniometry and digital image analysis, *Vet Surg* 34:463-468, 2005.



11

Canine Locomotion Analysis

Robert L. Gillette and T. Craig Angle

It is important to understand the internal and external factors of locomotion. A complete clinical assessment and rehabilitation program depend on an in-depth knowledge of the biomechanics of locomotion. Biomechanics is the study of the mechanics of a biological system and includes kinematic and kinetic principles. Kinematics refers to motion, whereas kinetics involves the forces associated with motion. Locomotion has been compared with a symphony orchestra in that “all parts must blend into a harmonious pattern—from the gentle sway of the head and tail for balance to the coordinated efforts of each limb and muscle to accomplish its special function. Conversely, also like an orchestration, if all movements are not attuned to the whole, a major fault should be evident.”¹

Locomotion

Previous biomechanical research has enhanced our understanding of canine locomotion and thus warrants an in-depth description.² The two primary types of locomotion are gait and nonrepetitive motions. Gait is a series of repetitive motions that include walk, trot, pace, gallop, and swim. Nonrepetitive motions do not occur in sequence and include single events such as jumping, movement initiation, and sitting. Nonrepetitive motions have received some focused research attention and are important to understand for clinical evaluations because forces and structural stresses can be at their greatest levels during these motions. By understanding the dynamics of gait and nonrepetitive motions, clinicians can better diagnose and treat injuries and diseases.

Locomotion requires a synergistic effort of the bones, joints, and neuromuscular systems to move, stop, and stabilize the body in the presence of sensory input. Locomotion depends on sensory input, and the sensory input determines how various static and dynamic responsibilities of the musculoskeletal systems are assigned. The musculoskeletal system is constantly distributing body weight, absorbing forces, maintaining posture, maintaining balance, and transferring the center of gravity over the base of support. The ability to properly carry out these static and dynamic responsibilities is what produces efficient and fluid locomotion. Any dysfunction in one joint or segment

of the skeletal system or any disruption of the neuromuscular system may result in an inability to carry out certain dynamic and static responsibilities. Asymmetries may develop between the angular displacements of joints, which produce discrepancies in timing and displacement of the limbs. This may lead to inefficient locomotion and/or lameness.

The skeletal segments and joints of the quadruped are repetitively accelerated, decelerated, and loaded with various force magnitudes. Depending on the geometry of the articular surface and the tissues surrounding a joint, and the load involved, each joint moves through a specific range of motion. There is a coordinated activation of neuromuscular components to rhythmically act on skeletal segments to produce fluid and symmetric motion about each joint. Neuromuscular components must produce concentric, eccentric, and isometric muscle contractions in a synergistic fashion. These muscle contractions must also be at optimal lines of action to apply forces to skeletal structures for efficient propulsion and force absorption. Skeletal structure position is directly influenced by the neuromuscular system and must maintain optimal joint angles and joint velocities to facilitate locomotion. Any alteration in motion of one joint by external devices, disease, injury, inflammation, or surgery may disrupt the flow and fluidity of locomotion.³ Also, joint motion can be indirectly altered psychologically by stimuli such as pain. All of the aforementioned alterations can be present in any quadruped, such as the dog.

Locomotion can be affected by the internal factors previously mentioned and by certain external factors. First, movement can be affected by the textural properties (e.g., vegetation and soil content) of the surface. A hard, soft, smooth, rough, and/or slippery surface affects movement by influencing traction and sensory input. Second, the angle or incline of a surface and the direction (i.e., up or down the incline) in which the dog is traveling affects movement and locomotion. Third, the curvature of a path such as running around a turn can directly influence the forces experienced by the dog. A curvilinear path subjects the dog to centrifugal and centripetal forces that the dog must withstand and compensate so as to stay balanced and continue with a sound gait. Fourth, a fluid environment can

affect movement by applying resistance to motion and buoyancy on the body. Finally, on a treadmill, the base of support passes under the center of gravity, whereas the center of gravity passes over the base of support overground. The specific way in which a treadmill affects neuromuscular patterns has not been studied, but its opposition in relation to support during overground locomotion is worth pointing out. Thus, there are both internal and external factors to consider when evaluating locomotion.

Gait

The gait describes a particular series of joint, segment, and whole-body movements used for locomotion. The temporospatial dynamics of each gait determine its type such as a swimming stroke,⁴ walk, trot, pace, or gallop. The temporospatial dynamics of a gait can be influenced by surface properties, surface incline, ground movement (i.e., treadmill), submersion in water, and whether or not the dog is traveling in a linear or curvilinear path.

A gait is made up of a series of repeated strides. Within a stride, each limb goes through a step cycle. A step cycle includes a stance phase and a swing phase (Figure 11-1). The stance phase is the period in which the foot is in contact with the ground. The first part of the stance phase occurs when the braking forces take place as a result of surface contact. This is followed by the second part of the stance phase in which there is a period of propulsion. The swing phase is the period in which the foot is in the air.

is divided into three parts. The leg first swings backward as a result of the propulsive action; the muscles then swing it forward for locomotion, and then bring it backward and down as it returns to the ground. Each one of these phases applies different forces on the limb's bony structure and musculature. Variations in these phases help the clinician when diagnosing lameness.

A stride is defined as the cycle of body movements that begins with the contact of one foot and ends when that foot again contacts the ground. There are various forms of gait that are defined by how the body moves during each stride.^{5,6} Symmetric gaits include the walk, trot, and pace. The trot is a two-beat diagonal gait in which the movements of one stride have the diagonal fore and hind limbs in support followed by the other two limbs in support (Figure 11-2). In this gait, the right side and left side of the body move symmetrically, or the two sides mirror each other.

Asymmetric gaits include the rack, gallop or canter. In asymmetric movement the sides move differently. The gallop is the gait typically used by dogs that are running at high speeds.⁷ It has two support phases and two flight phases in each stride (Figure 11-3). The gallop can have either a right or a left lead. In left-lead dogs the feet contact the surface in a clockwise fashion. The left front leg comes into contact first, followed by the right front foot in the front support phase. The dog then leaves the ground in the front flight phase. The next foot to make contact is the right rear foot, followed by the left rear foot in the rear support

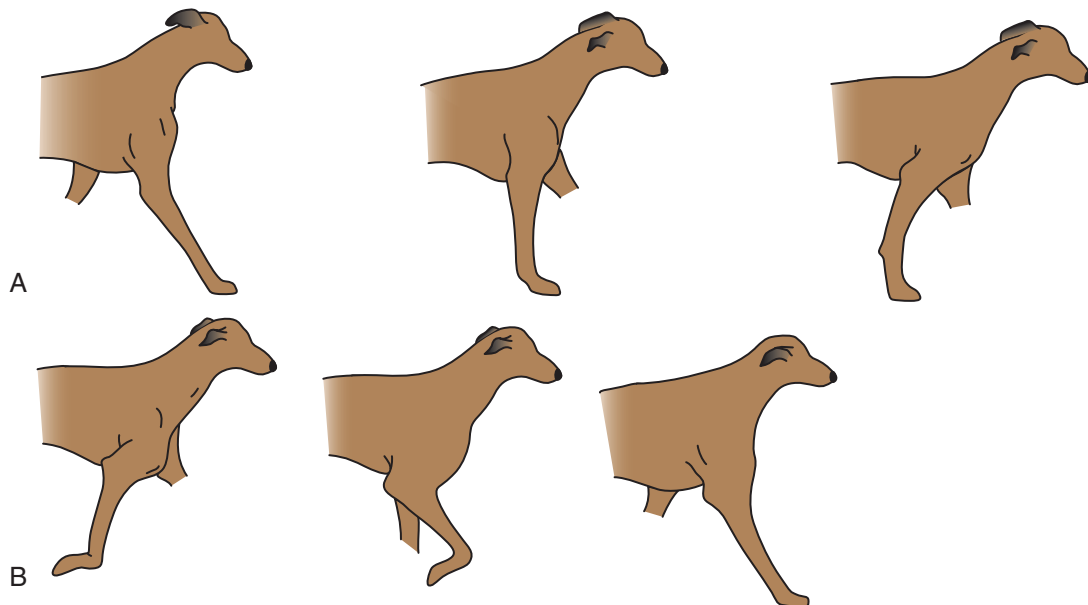


Figure 11-1 A, The stance phase. Braking forces occur at initial impact and continue until the center of gravity transfers over the fulcrum. Propulsive forces predominate for the remainder of the stance phase until the foot leaves the surface. B, The swing phase. Initially the leg swings caudally after the foot leaves the ground. It then swings cranially, until it begins the last caudal swing, which ends when the foot touches the surface.

phase. The dog then propels forward, leaving the ground a second time in the rear flight phase, after which the sequence is again repeated. A right-lead dog goes through the same sequence but contact is in a counterclockwise rotation. This gait is described as a double suspension-rotary gallop. The canter is slower than the gallop and is a three-beat gait, characterized by a “lead” forelimb, with two limbs moving separately and two as a diagonal pair. The pattern of footfalls with a dog cantering in a left lead is right hind, left hind and right fore together, followed by the left forelimb reaching furthest forward. The opposite series of footfalls would occur when a dog is cantering on the right lead. There is only one flight phase in the canter when all four limbs are off the ground, and it occurs just after the forelimbs leave the ground. The canter, sometimes called a lope, is an endurance gait and is sometimes used by sled dogs. Some of the larger dogs that have a wider body structure, such as the Rottweiler or Saint Bernard, use the canter as their preferred running gait.

Nonrepetitive Motions

Little research exists evaluating nonrepetitive motions. Therefore there have been no scientifically accepted definitions of the movements (e.g., jumping and movement initiation). One paper describes jumping⁸ and another describes movement initiation.⁹ Understanding these movements is important; for example, movement initiation

is the movement that is used to begin all gaits. The forces of nonrepetitive motions are often higher than most normal gaits such as the trot, and it is possible that noncontact injuries may occur during nonrepetitive motions. The motions yield not only high forces but also high rates of force application to musculoskeletal tissues. Nonrepetitive motions generally involve a wide range of joint motion and segmental alignments with high force outputs and impacts. From a clinical standpoint, it is important to understand these movements because they can cause injuries or can be used to reveal injuries that might not be noted on a routine clinical exam involving a trot.

Structural Definitions

Movement is balanced throughout the whole body in a normal healthy animal. In the neuromusculoskeletal system, the two sides of the body mirror each other and are symmetric. An understanding of body symmetry is the key to analyzing gait abnormalities and lameness. Whether palpating the dog during a physical examination or observing the dog’s movement, the right side of the body should be symmetric to the left side.

The dog’s structure consists of axial and appendicular sections, which are divided into segments when analyzing motion. The axial vertebral column is made up of many joints and is divided into anatomic segments. The cranial segment is the head, followed caudally by the neck (cervical), thoracic, abdominal (lumbosacral), and tail segments. The appendicular segments are the front legs and the back legs. These are subdivided into smaller segments by the joints: shoulder, elbow, carpus, hip, stifle, tarsus, and phalanges. Locomotion as a whole is a result of the individual movements of these segments. How each of the segments and joints is used is determined by two factors. The first factor is the relationship of the dog’s center of gravity to the surface, and the second is the type of movement that the body is performing.

Conformation

A major factor that affects how the body moves and receives forces is conformation. Conformation is the symmetry, size, and shape of various body regions relative to each other or in general appearance. Improper body design predisposes a structure to injury. Gait analysis reveals certain characteristics of a dog’s conformation. This plays an important role in breeding programs. Sound movement contributes to the health and normal life span of all dogs.¹⁰ Locomotion can vary for different breeds,⁶ but in a

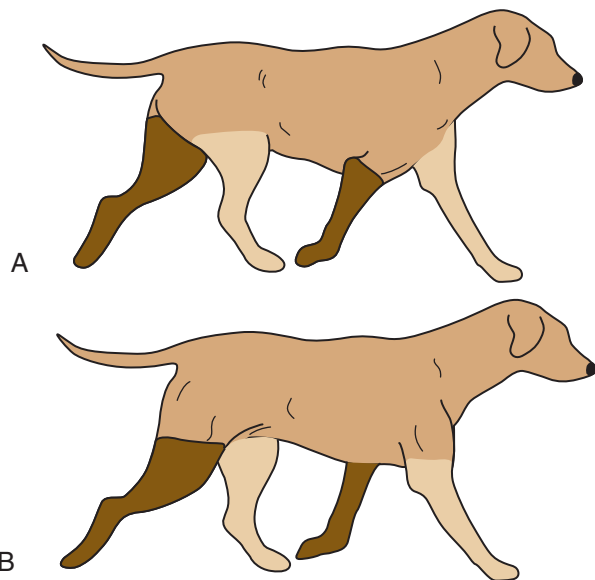


Figure 11-2 The trot. The limbs in light brown are in the stance phase, while the dark brown limbs are in the swing phase.



Figure 11-3 The gallop, with the dog in the right lead.

symmetric gait, like the trot, the movements of one side should still mirror the movements of the other side. A conformational abnormality usually creates an abnormal gait. Any abnormal gait should first be assessed for a conformational cause. Lameness of this type may not be related to pain but to a structural fault. Conformational abnormalities that can lead to musculoskeletal problems should be eliminated from breeding programs. If incorrect movement, or lameness, is not a result of compensation because of conformational abnormalities, other factors must be influencing the movement, such as an injury.

Center of Gravity

In the dog, 60% of the dog's weight is carried by the front limbs and 40% of the weight is carried by the back limbs while standing.⁷ This results in the center of gravity (COG) being at the mid-chest level behind the scapula in the dog. The placement of the COG results in the front legs being associated more with braking forces, whereas the rear legs are more associated with propulsive forces on a level surface. Moving uphill shifts the balance of forces to the back legs and moving downhill shifts the forces to the front legs. An uneven or varying surface produces a variety of force alterations that depend on the body's position. For this reason, all general analysis should be performed on a level, even surface that provides proper traction.

Impact Forces

How a dog runs determines how the body receives the forces of surface impact. A dog that uses the double-suspension rotary gallop has high-impact forces on the front legs as they contact the ground. This is because as the limbs contact the ground, the entire weight of the body as well as the added braking and propulsion forces must be absorbed by the limbs. These forces have a greater effect on the lead leg than the non-lead leg. In the breeds that use the canter when they run, the impact forces are distributed differently. The forces of impact are distributed more evenly between the front and back legs.

Locomotion Analysis

In the past two decades, technologic advances in computer-assisted gait analysis have aided our ability to quantitatively define temporospatial gait characteristics. This has assisted researchers and clinicians in gaining a better understanding of canine locomotion. Subjective evaluation of canine gait has been used for many years. However, our ability to perceive minute details during the gait cycle can be very difficult, and in some respects, impossible, even for the most talented gait specialist. During a subjective evaluation, a clinician is only able to perceive a few kinematic variables at a time, but a kinematic or kinetic analysis system can capture, analyze, and store hundreds of observations per second. In addition, clinicians have the

ability to view an animal in motion, but cannot determine the forces involved in gait and identify specific neuromuscular activity; therefore, we must employ kinematic and kinetic analysis tools for a more complete description of gait.¹¹

Subjective Kinematic Analysis

Kinematic analysis is one of the oldest methods to evaluate movement of the musculoskeletal system. In 1888, Muybridge used stroboscopic photography to show the stride of racing greyhounds and hound dogs running at various speeds.¹² Subjective kinematic analysis is the most common diagnostic tool to assess lameness. The clinician evaluates the dog by observing gait and/or nonrepetitive motions and looks for abnormal kinematics. When analyzing movement, a clinician should define the mechanical system under analysis. The mechanical system can be the dog as a whole or a specific joint or limb.

To describe the components that make up a gait or motion, the movements of the joints, individual segments, and body as a whole are observed. These are interpreted in terms of time, distance (i.e., degrees or meters), speed, consistency, and accuracy. The trot, which is a symmetric gait, provides a good visual presentation of movement for the clinician to diagnose lameness. If a gait abnormality is present, the movements of one side are different from the other side.

The observations should be conducted before any physical palpation. First observe the animal while it is at rest, looking for conformational abnormalities or abnormal stances. For example, does the dog seem to hold one leg up or put most of its body weight on a particular leg? After these observations are noted the animal is analyzed while moving. If able, trot the patient in a straight line toward and away from the clinician. Next it should be assessed moving in a straight line from the right side and then the left side. Then it should be observed moving in a circle, clockwise then counterclockwise.

Movements of the body as a whole should be assessed for any major gait deficits first. Then the step cycle of each leg should be evaluated to determine which phase of the cycle is deficient. If the lameness appears at foot contact only, or the stance phase is shortened, the lameness can be associated with impact or support. An example of a structure that presents with this type of gait abnormality is the paw. The tissues in this area are the pads and skin, bones, ligaments, joint capsule, and tendons. The entire limb should be examined using physical palpation to locate which tissues are involved.

Most obvious abnormalities can be detected with subjective gait analysis. A dog with a lesion causing severe sharp constant pain carries the limb and keeps the weight off it. A dull aching pain produces a limp during gait. A lesion that produces a small pain that occurs in certain phases of locomotion allows the dog to adjust its gait for

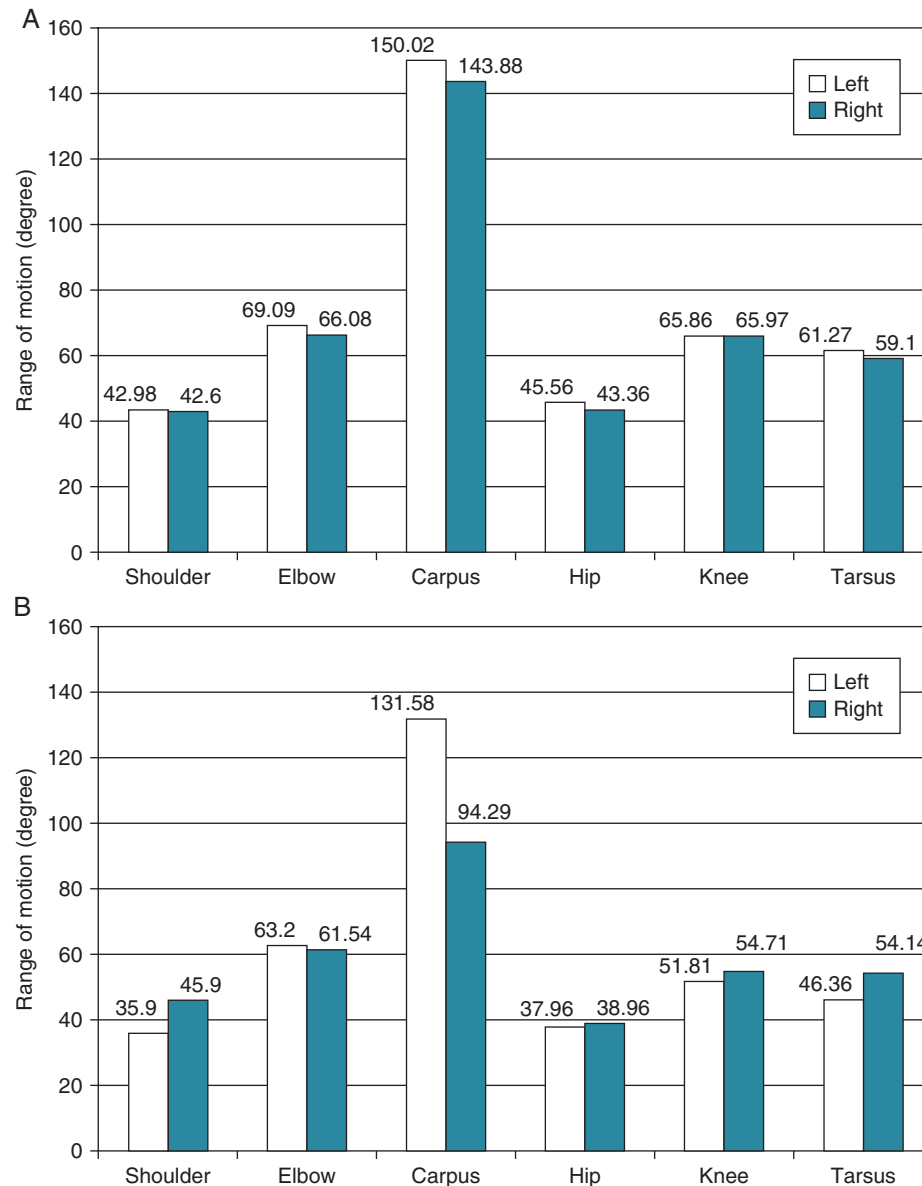


Figure 11-4 **A**, The chart shows actual range-of-motion values of sound dogs during a trot. **B**, The chart shows the actual range-of-motion values of a dog with restricted carpal movement.

relief. The quadruped has the ability to minimize pain by altering movement in such a way that the abnormality may be unnoticeable.¹³ This altered gait can lead to subsequent compensatory problems.

The normal trot is a balance of movement by the whole body (Figure 11-4, A). Minor pain can create an imbalance in the system. For example, synovitis of the right carpal joint affects not only the joint of injury, but may also affect the left hip and left tarsus (Figure 11-4, B). An altered gait affects how the forces of movement act on the musculo-skeletal structure. In addition to appendicular changes, the axial skeleton is affected. It is the frame through which this alteration is transferred. This can initiate two problems,

one biomechanical and the other psychological. Biomechanically, if a condition is untreated the compensatory movements can lead to other injuries in the back or limbs. Psychologically, chronic pain can lead to behavioral problems.

Quantitative Kinematic Analysis

Quantitative kinematic analysis quantifies the positions, velocities, accelerations, and angles of anatomic points, segments, and joints in space. Kinematic analytic systems track anatomic markers through different planes of space and time and reveal their respective x, y, and z positional values. From the positional values, software programs can

use mathematical formulas to calculate linear or angular velocities and accelerations; therefore these systems assign numeric values to motion for quantitative gait analysis.¹¹ Some academic institutions currently have laboratories that can perform motion or gait analysis. These laboratories usually are equipped with a three-dimensional motion analysis system used for research or clinical analysis. Information about the animal's motion can assist in lameness diagnosis, assess rehabilitation progress, and possibly assess athletic potential. Veterinary medicine is just beginning to evaluate how motion analysis can raise the standards of clinical care. This is a rapidly growing field that will extend the boundaries of knowledge of patients' neuromusculoskeletal systems.

Traditionally these systems are too costly for use in most general practices and are relatively time-consuming to operate. An alternate method is two-dimensional gait analysis.¹⁴ Reflective markers are placed over anatomic points (Figure 11-5). The subject is then trotted with its right side to the camera and then with its left side to the camera. The movement is filmed by a video camera system. The video clip is entered into a computer and each of the reflective markers are digitized and recorded. The points generally digitized are the lateral aspect of the distal fifth digit of the front paw, the carpus, the elbow, the acromion/

greater tubercle of the humerus, the dorsal aspect of the scapular spine, the lateral aspect of the distal fifth digit of the rear paw, the tarsus, the stifle, the greater trochanter of the femur, and the dorsal iliac crest. Both sides of the body are digitized. These data points are used to quantify segmental and linear movement (Figure 11-6).

The clinician can evaluate the data from both sides for asymmetry. When assessing gait and joint movement at the trot, any differences between the two sides may be used to evaluate lameness. When using proper methodology, two-dimensional gait analysis is valid and reliable. It requires only one video camera. It is important to only evaluate the gait cycle directly in front of the camera. Two-dimensional analysis systems are more economical than three-dimensional systems, allowing more practitioners an opportunity to use this tool.

In addition, filming outside of the laboratory is possible. Videotapes may then be evaluated by a laboratory, allowing further benefits of computer-assisted video gait analysis.

Future kinematic studies should include obtaining more information on different breeds. This may help define conformation standards for breeding purposes. Increasing the number of dogs analyzed with diagnosed musculoskeletal pathologies helps to determine parameters that are associated with those specific etiologies. After these parameters are determined, these methods may be used to quantify response to treatment.

Professionals, trainers, and owners should be educated on the use of computer-assisted video gait analysis. The more information that is distributed the greater the potential to benefit from the knowledge gained about animal locomotion.

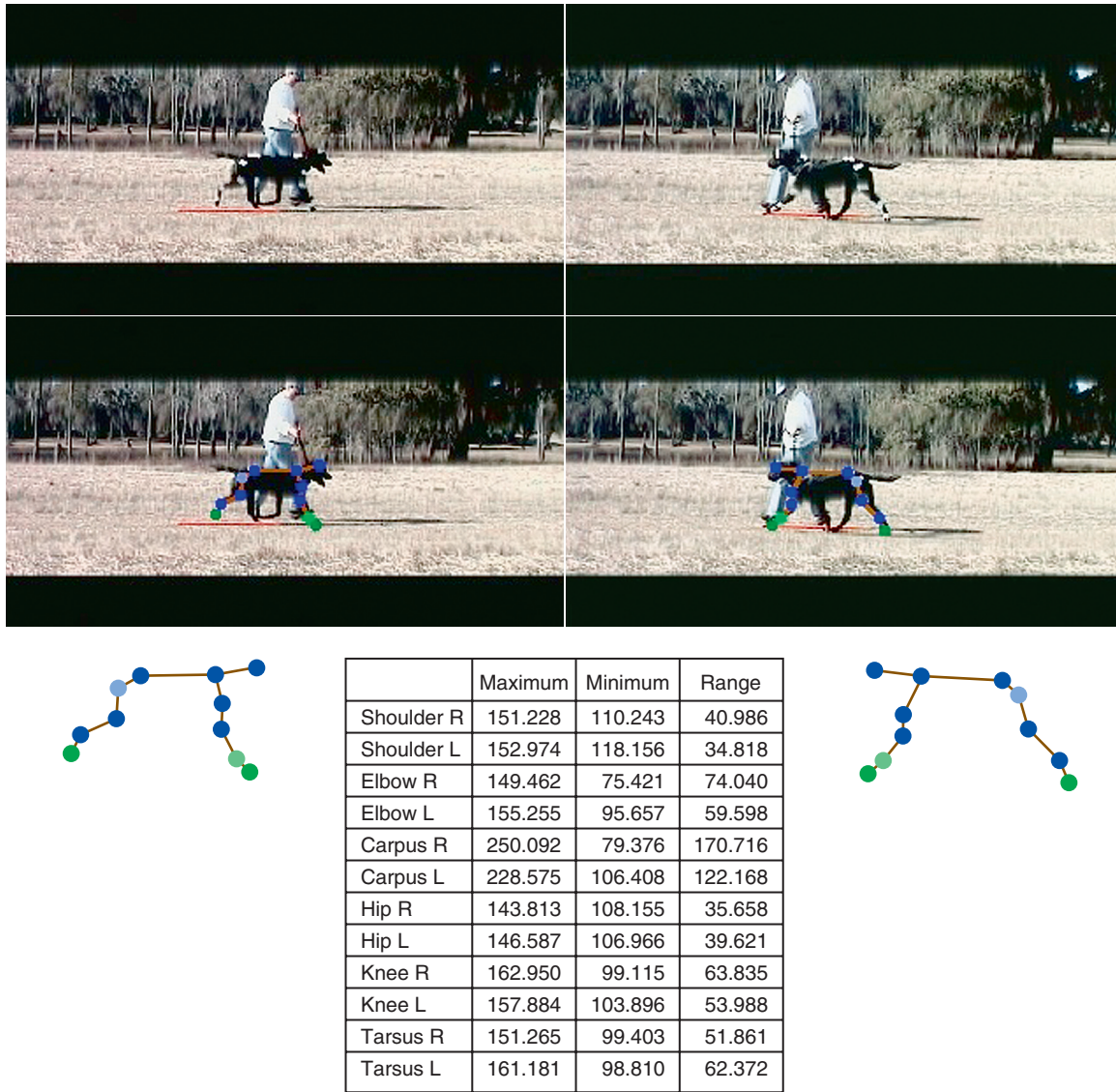
Kinetic Analysis

Kinetic (force) analysis quantifies the forces of locomotion by using measurement systems to evaluate kinetic variables, including peak vertical, braking, propulsive, and horizontal forces; vertical, braking, and propulsive impulses; strain within various tissues; rates of loading; temporal gait characteristics; and pressure distributions of the paw. There are many different methods and systems used to evaluate kinetic variables. Most of the systems used are ground-based (e.g., force plate and force mats) kinetic systems and measure orthogonal ground reaction forces (i.e., mediolateral force, craniocaudal force, and vertical force) resulting from paw impact during the stance phase of gait. Other systems, such as strain gauges, measure strain of different tissues when stress is applied during gait.¹¹

The force plate can be used to measure ground reaction forces (GRFs) exerted on the limbs. The forces are measured in three orthogonal components: the vertical (z) plane, cranial-caudal (y) plane, and medial-lateral (x) plane (Figure 11-7). The force plate is usually embedded in a pathway for the test subjects to pass over. As the paw



Figure 11-5 Reflective markers placed on the dog before filming. Markers are seen during locomotion.



A

Figure 11-6 Video capture of gait from a dog using a 2-D video capture system. Kinematic data may be acquired from dogs working in the field.

Continued

comes into contact with the plate, GRFs are measured. The plate is connected to a computer that acquires the data for analysis. The data can be used to assess how the forces of the body are transferred through the limb. For analysis, the data from one limb can be compared with data collected over time or other limb data from the subject.

The measurements are made and read as force exerted over time. The contact time depends on the velocity of the subject. As the velocity increases, the contact time decreases. For example, the paw contact time for the walk is longer than the contact time in the trot. The walk has a biphasic z -force for each limb strike, which is similar to that of humans. The first peak is associated with the braking forces that occur at initial impact, and the second peak is a result of the propulsive forces. This is also seen in the y plane, with an initial negative force from the braking phase,

and then a positive force from propulsion. The x -forces play a minimal role in gait evaluation. The biphasic z -force is not seen in the trot, presumably because of decreased contact time and the lack of a heel strike and toe-off in the dog. There is a single loading peak for each limb strike in the z -plane (Figure 11-8, A). The y -forces are biphasic, with braking and propulsive components (Figure 11-8, B). The contact time of the walk is longer than that of the trot, but the z -force of the trot is twice the force of the walk, as a result of the increase in velocity, and the limb support differences between the walk and the trot.

A technique has been reported for measuring ground contact pressure on individual paw pads of dogs.¹⁵ The force plate measures the ground reaction forces transferred through the entire paw. To assess how the paw receives the GRFs, each individual digit may be assessed. This is done

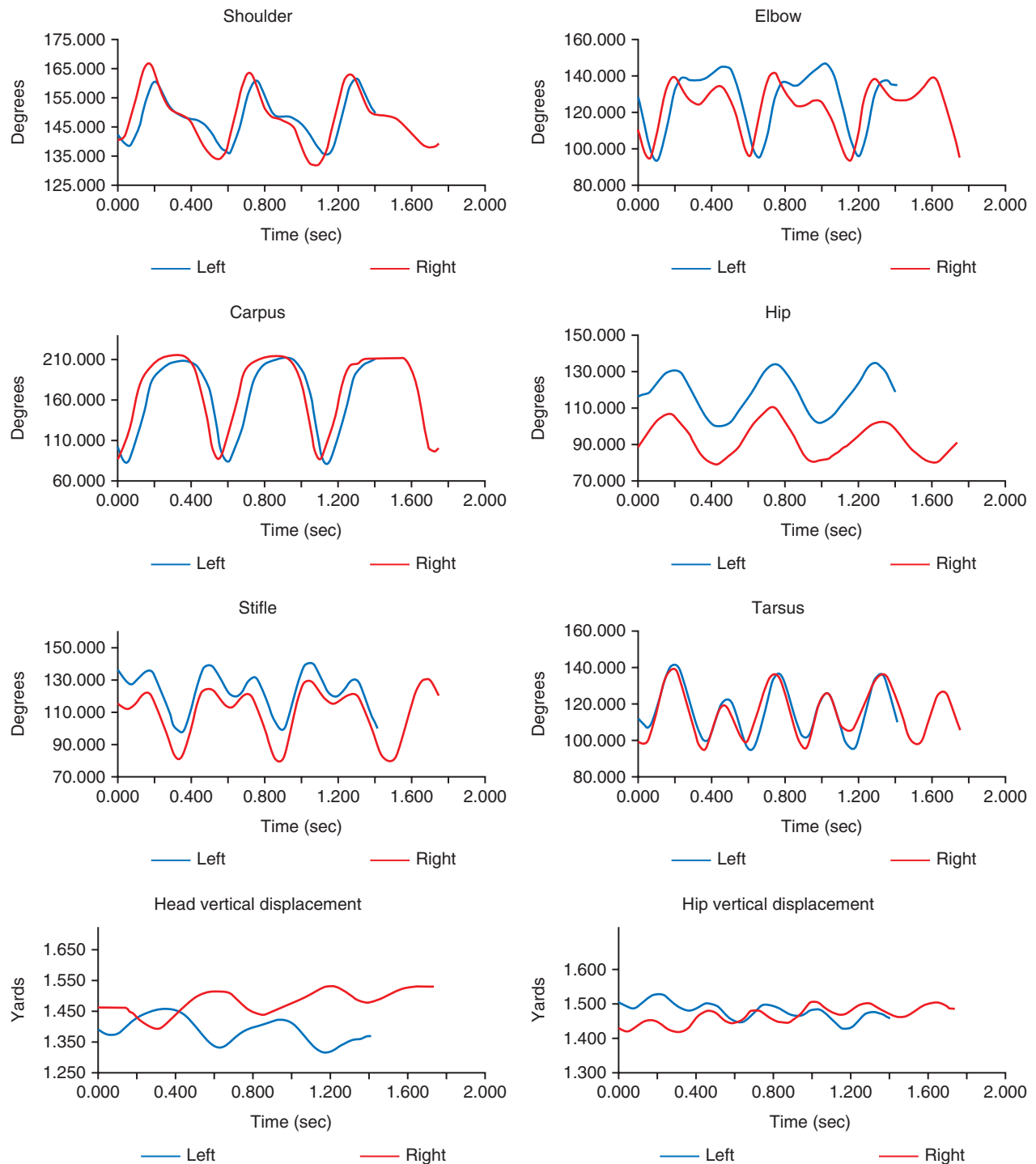


Figure 11-6, cont'd Symmetry of Joint Motion may be evaluated with Kinematic data acquisition systems. For example, the right elbow, tarsus, and stifle joints have reduced ROM compared to the left side.

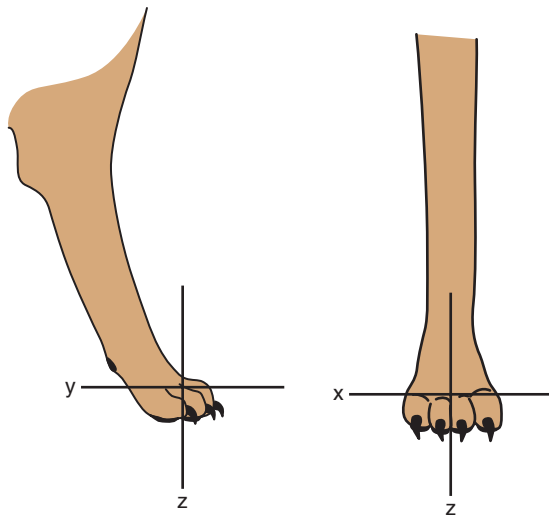


Figure 11-7 Ground reaction forces act on the foot in three orthogonal planes (x, y, and z).

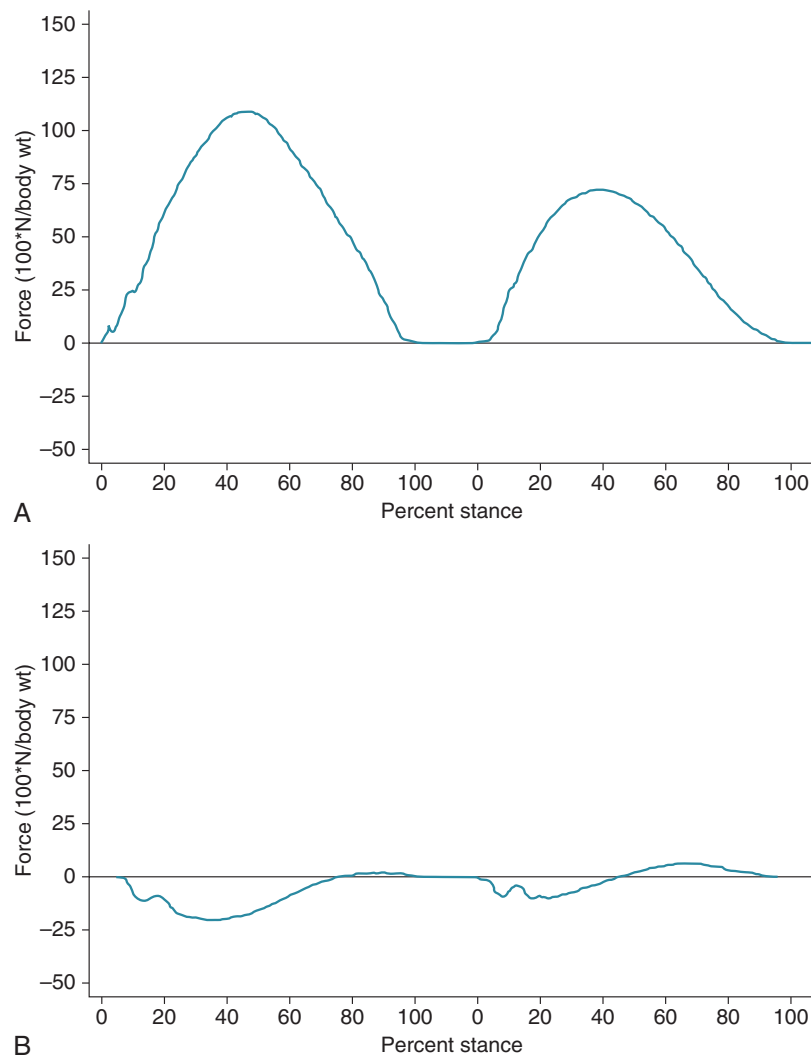


Figure 11-8 **A**, Force plate data showing measurement of the z force during surface contact at the trot. **B**, Measurement of y force taken during the trot.

by attaching a force-sensing resistor to each individual weight-bearing pad. One unexpected finding in one study was that the greatest pressures were on digital pads 3 and 5, and pads 2 and 4 had the least amount of pressure.¹⁵ It had previously been thought that adjacent digits 3 and 4 were the pads of greatest pressure. This technique of measurement can be used for evaluating bandaging methods, prosthetic devices, and surgical manipulations for alleviating pressures.

Pressure Measurement Systems

Pressure mat systems are made up of many pressure sensors. When a dog walks over the mat, visual graphics of pressure distributions of the paw appear on the computer screen. These systems quantify high and low pressure areas of the paw, relative or actual vertical forces involved in the stance phase, and temporal characteristics of the stance phase. Pressure mats come in different sizes. Pressure mats in combination with a laptop computer provide a convenient portable kinetic data collection system.¹¹

Summary

Locomotion analysis is a helpful analytic tool for the clinician, technician, handler, or owner. It is a good indicator of the status of the musculoskeletal system. The evaluator should have a good understanding of the individual components of locomotion, anatomy, conformation, and the influence of external factors. It is important to remember that center of gravity and conformation influence how ground reaction forces interact with the body. New technologies have advanced our ability to evaluate motion and forces. Veterinarians, physical therapists, breeders, owners, and trainers all can potentially benefit from the recent advances in locomotion analysis. Improved capabilities for data collection, storage, and analysis, in combination with advances in personal computer technology, have provided an opportunity for these analytic techniques to be used in the respective areas of interest. The more information that is gathered

and distributed, the greater the potential to benefit from the knowledge gained regarding animal locomotion.

REFERENCES

1. Roy WE: Examination of the canine locomotor system, *Vet Clin North Am* 53:70, 1971.
2. Gillette RL, Angle TC: Recent advancements in canine locomotor analysis: a review, *Vet J* 178:165-176, 2008.
3. Eward C, Gillette RL, Eward W: Effects of unilaterally restricted carpal range of motion in kinematic gait analysis of the dog, *Vet Comp Orthop Traumatol* 16:58-163, 2003.
4. Marsolis GS, McLean S, Derrick T et al: Kinematic analysis of the hind limb during swimming and walking in healthy dogs and dogs with surgically corrected cranial cruciate ligament rupture, *J Am Vet Med Assoc* 222:739-743, 2003.
5. Grogan JW: The gaits of horses, *J Am Vet Med Assoc* 118:112-117, 1951.
6. Hildebrand M: Symmetrical gaits of dogs in relation to body build, *J Morphol* 124:353-360, 1968.
7. Zebas CJ, Gillette RL, Hailey RL et al: Kinematic descriptors of the running gait in the greyhound athlete. In Marshall RN, Wood GA, Elliott BC, et al, eds: *XIIIth International Conference on Biomechanics*, Perth, Australia: University of Western Australia, 469-470, 1991.
8. Alexander RM: The mechanics of jumping by a dog (Canis familiaris), *J Zool, Lond* 173:549-573, 1974.
9. Angle TC, Gillette RL, Wiemar W: (Dissertation) *Kinematic analysis of two natural surfaces on the canine sprint start*, 2009.
10. Elliot RP: *The new dogsteps*, New York, 1983, Howell Book House.
11. Gillette RL, Angle TC: Recent advancements in canine locomotor analysis: a review, *Vet J* 178:165-176, 2008.
12. Muybridge E: *Animals in motion*, New York, 1957, Dover.
13. Nunamaker DM, Blauner PD: Normal and abnormal gait. In Newton CD, Nunamaker DM, eds: *Textbook of small animal orthopaedics*, Ithaca, NY, 1985, IVIS.
14. Gillette RL: (thesis) *Validation of a two-dimensional analysis of limb symmetry in the trot of Labrador Retrievers*, 1998.
15. Rumph PF, Marghitu DB, Gillette RL et al: (abstract) *A technique for measuring ground contact pressure on individual paw pads of dogs*, Proceedings of the Annual Meeting of the American Association of Veterinary Anatomists, 1998.



12 The Physical Rehabilitation Evaluation

*David Levine, Denis J. Marcellin-Little, Marti Drum,
and Cassy Englert*

Physical therapy is a profession with an established scientific basis in humans and companion animals. It has numerous clinical applications in the restoration, maintenance, and promotion of optimal physical function.¹ It is beneficial in helping people recover from anterior cruciate ligament reconstruction, fracture stabilization, joint surgery, spinal surgery, and many other injuries or diseases.²⁻⁵ It also improves function in a variety of patients with soft tissue injuries, osteoarthritis, total joint arthroplasty, stroke, and chronic lower back pain throughout their lives.⁶⁻¹⁰

In providing physical therapy the goal is to restore, maintain, and promote optimal function, optimal fitness, wellness, and quality of life as they relate to movement disorders and health. In dogs, this may include treating patients during their recovery from orthopedic surgical procedures, such as femoral head ostectomy, monitoring weight loss programs, strengthening specific muscle groups, and helping to manage chronic conditions, such as osteoarthritis, or progressive conditions, such as degenerative myelopathy. A major emphasis is to prevent or minimize the onset, clinical signs, and progression of impairments, functional limitations, and disabilities that may result from diseases, disorders, conditions, and injuries. Examples in people include designing and delivering treatment programs for patients with problems such as pneumonia, multiple sclerosis, diabetes, cerebral palsy, lower back pain, or frozen shoulders.^{1,11-14}

Treatment Philosophy

Physical therapists use a variety of treatment interventions such as stretching, targeted massage, passive range of motion, and joint mobilization. They also use electrical and thermal modalities and therapeutic exercises to help patients reach their goals. These treatments work synergistically to effectively achieve the therapeutic goals. When designing a treatment plan, the therapist should be aware of the scientific evidence supporting the use of each modality and exercise for the problems being treated. For example, when treating postoperative edema, ice has been proved to be beneficial,¹⁵ whereas certain other therapies are unproven. The therapist should integrate the individual

treatment plan with established perioperative and postoperative pain management protocols. Although the clinical signs present in many dogs with orthopedic or neurologic problems may improve over time, such as after fracture repair, a well-designed physical rehabilitation program may accelerate the recovery, may prevent permanent disability, and may help prevent future re-injuries.

Patients with movement disorders, weakness, pain, and limited endurance are candidates for physical rehabilitation. Examples of conditions include dogs recovering from orthopedic or other surgeries, dogs with osteoarthritis, tendonitis, or other soft tissue injuries. After a medical diagnosis is available, the therapist evaluates several aspects of the patient's health, particularly the health of the cardio-pulmonary, neurologic, orthopedic, and integumentary systems. The more specific the medical diagnosis, the more directed will be the care. For example, the medical diagnosis for a patient may be osteoarthritis of the elbow and the physical rehabilitation diagnosis for that patient may be limited flexion and extension with cranial and caudal joint capsule tightness. These factors may limit function in terms of gait; improving elbow range of motion through specific treatment interventions may improve the functional status of the patient. In the practice of physical therapy for humans, the areas evaluated include aerobic capacity, balance, arousal, cognition, environmental barriers, ergonomics, posture, gait, pain, range of motion, prosthetic requirements, and assistive and supportive devices. Most of these parameters may be evaluated in dogs.

Physical rehabilitation in veterinary medicine follows the same principles. The therapist collects functional information by evaluating the dog's physical fitness, orthopedic, and neurologic health. This may be done in conjunction with or after the veterinarian's evaluations. There is overlap in these evaluations, but although the veterinarian evaluates the patient to obtain a medical diagnosis and to prescribe medical or surgical treatment plans, the therapist evaluates the patient to create a physical rehabilitation treatment plan. This evaluation includes the assessment of muscle mass, joint motion, joint stability, and pain. For example, loss of range of motion may be present with elbow dysplasia, whereas loss of sensation and muscle atrophy may be observed with radial nerve injuries.

Physical rehabilitation plays an important role in prevention of and recovery from injury. Educating owners in proper warm-up and cool-down techniques, for example, may help prevent orthopedic injuries during training or competition. Owners may have a substantial emotional and financial investment in their dogs and are generally very willing to assist in carrying out rehabilitation programs when properly instructed.

Physical Rehabilitation Evaluation

The physical rehabilitation evaluation involves the evaluation of active motion during various gaits and while performing specific activities, such as stair climbing. The dog should be evaluated at a walk, trot, and gallop if the patient is able and it is safe to do so. The evaluation also includes the assessment of function at rest, including posture, balance, and strength. Balance may be assessed by performing various perturbation maneuvers such as pushing the dog off balance and assessing the dog's response. Balance is further tested by evaluating overall coordination, gait, and whether or not the patient falls. Proprioception during ambulation is tested by evaluating toe dragging, knuckling, awkward stepping, and limb carriage and placement. Strength is tested by evaluating muscle mass and the ability to perform normal activities. For example, hip extensor strength may be assessed by observing the dog when it rises. The evaluation continues with assessment of

pain and motion of joints. The therapist evaluates discomfort and the condition of injured and abnormal tissues, such as excessive laxity or restriction of motion as a result of scar formation (Table 12-1). The therapist collects this information to understand how these limitations affect function and how they may be targeted during therapy.

To help determine the cause of articular, muscle, or connective tissue restrictions in motion, the examiner assesses the *end feel*. The end feel is the sensation imparted to the examiner's hands at the end of the range of motion of the tissue being examined (Figure 12-1).¹⁶ The normal end feel for most joints is imparted by the joint capsule and is a reflection of the slight elasticity of that capsule. Abnormal end feels are described in Table 12-2.

A typical rehabilitation evaluation form (Figure 12-2) allows the evaluator to easily document the pertinent parts of the evaluation and be able to refer back to them as treatment progresses. It may also be used as a form of communication between the rehabilitation practitioner and the referring veterinarian (RDVM).

Rehabilitation Candidates

All dogs that have neurologic or orthopedic problems are candidates for rehabilitation, particularly hunting and working dogs and those dogs that perform strenuous physical activities, such as agility, racing, field trials, police activity, and Schutzhund. Rehabilitation also applies to

Table 12-1 Specific Tissue Assessments during the Physical Rehabilitation Evaluation

Tissue	Parameters	Anomalies	Potential Causes
Muscle	Mass/size	Atrophy	Denervation, disuse, contractures
		Hypertrophy	Increased training and conditioning
	Tone	Increased	Edema, hematomas, neoplasia
		Decreased	Spasm, guarding, denervation (upper motor neuron)
Tendon	Pain	Acute pain	Denervation (lower motor neuron)
		Chronic pain	Tear, myositis, spasm
	Tension	Acute pain	Tear, contracture, spasm, referred neurogenic pain
		Chronic pain	Tendonitis, strain
Joints	Motion	Increased	Tendonitis
		Loss	Adhesion, contracture
	Pain	Abnormal	Contracture, periarticular fibrosis or adhesions, effusion
		Acute pain	Subluxation, luxation
Ligaments	Stability	Chronic pain	Subluxation, luxation, infection
		Decreased	Osteoarthritis
	Pain	Increased	Sprain, rupture
		Acute pain	Adhesion, contracture
		Chronic pain	Sprain
			Adhesion

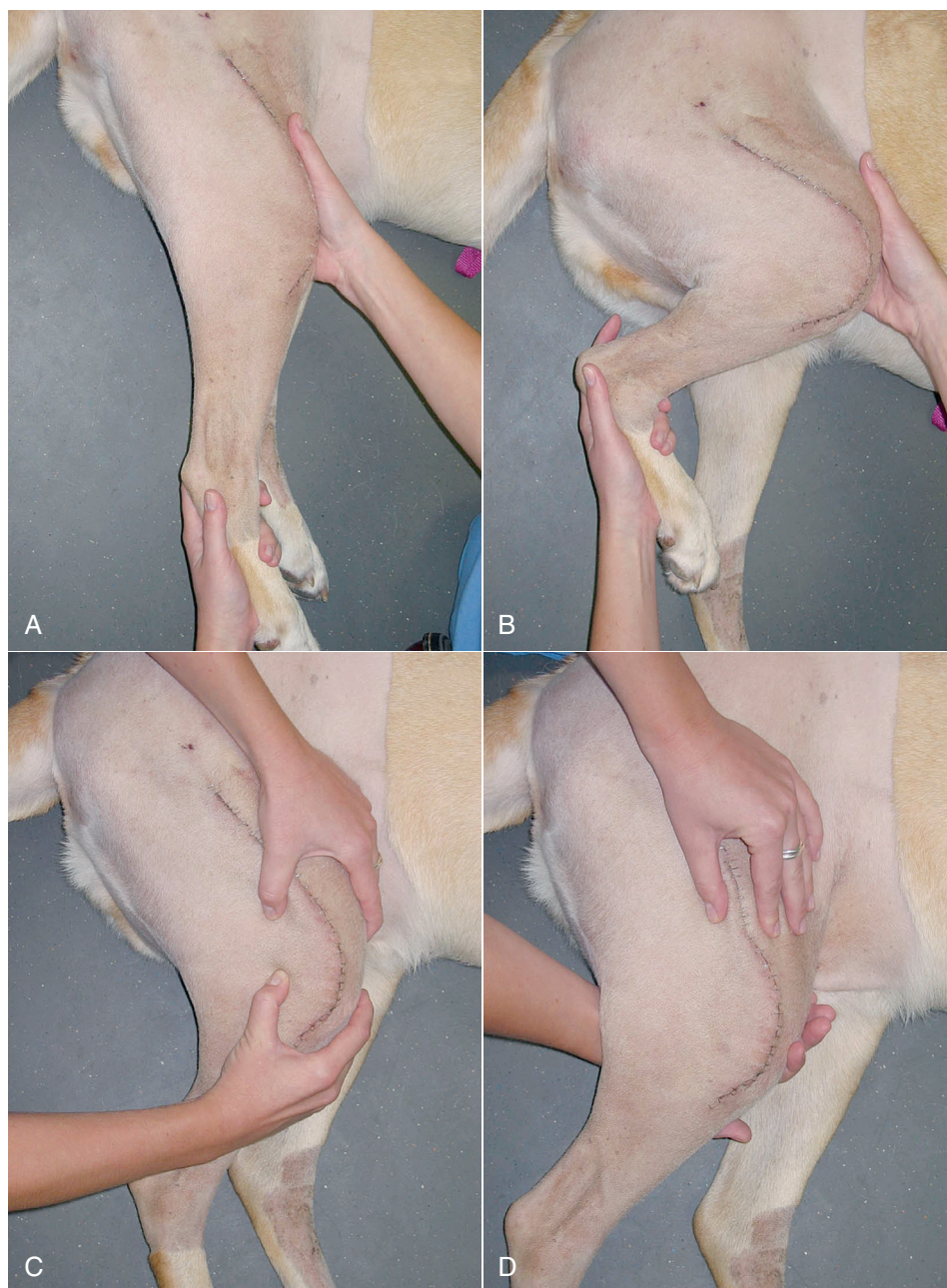


Figure 12-1 This Labrador retriever is recovering from an open surgical lavage for treatment of a large hematoma and infection after extra-capsular stabilization of a torn cranial cruciate ligament. A physical therapist evaluates range and end-feel with stifle extension (A) and flexion (B), the presence of cranial drawer (C), and swelling and adhesions of tissue planes (D) in the operated area.

dogs recovering from surgery. For example, range of motion exercises may be used after repair of a fracture of the distal femur in a puppy to prevent quadriceps tie-down. Also, a reconditioning program may be instituted after a major medical condition or abdominal surgery to help return the patient to its previous status. Some of the common conditions appropriate for rehabilitation include surgery for cranial cruciate ligament rupture, patellar luxation stabilization, femoral head and neck ostectomy (FHO),

various fractures with and without surgical repair, spinal disk decompression, neurologic conditions such as fibrocartilaginous embolic myelopathy, ligamentous instabilities, overuse injuries, muscular imbalances, chronic pain, and weight loss and general conditioning. Weight management is quickly becoming a common reason for referral to rehabilitation. Obesity in dogs can lead to many of the preceding orthopedic injuries, pain, and general health concerns. The rehabilitation practitioner commonly uses a

Table 12-2 Anomalies in End Feel Assessed during the Physical Rehabilitation Evaluation

Type of End Feel	Definition	Potential Causes	Example
Capsular	Slight elasticity	Normal joint	Normal shoulder extension
Firm capsular	Decreased elasticity	Periarticular fibrosis, adhesions	Loss of stifle extension following cranial cruciate ligament rupture
Springy	Increased bounce	Joint effusion, joint mouse	Always abnormal
Soft tissue approximation	Motion limited by soft tissues	May be normal or caused by swelling	Normal hip flexion
Empty	End of motion cannot be reached	Pain	Intraarticular fracture
Hard	Abrupt stop	Bone on bone contact, mature contractures	Chronic quadriceps contracture following distal femoral physal fracture

variety of therapeutic exercises, including an underwater treadmill to exercise obese or overweight dogs because the buoyancy of the water decreases joint stress while increasing the metabolic rate for successful weight loss (Figure 12-3). Rehabilitation for weight management works best when done in conjunction with the referring veterinarian's recommendations regarding diet and medication if necessary.

Developing a Physical Rehabilitation Plan of Care

Based on the results of the medical history and diagnosis, as well as the physical rehabilitation evaluation, a clear, evidenced-based plan of care is developed and implemented. In some instances, the plan of care is designed using evidence-based conclusions from human studies and having knowledge of the anticipated tissue responses to therapy in dogs. The therapist chooses the treatment plan that has the fastest, most predictable positive outcome on the patient. Acute postoperative edema, for example, may be treated using ice, bandaging, and pain-free passive range of motion. As the acute inflammation subsides, gradual resumption of activity may be combined with hot packs or ultrasound therapy. The plan of care is subsequently modified based on frequent reassessments. Patients may be treated as inpatients in the hospital or as outpatients in rehabilitation centers. The plan of care must be unique to the patient and must take into account all abnormal findings and other factors, including the severity of the anomalies, the age and disposition of the dog, the expectations for future performance, the urgency of the recovery, the available equipment and technical skills of the clinicians, and the cost of treatment. A typical plan of care may include a choice of thermal and electric modalities and specific exercises aimed at strengthening with gradual reintroduction of specific physical activities. The therapist

prescribes the specific type, intensity, duration, frequency, and progression of these exercises. Exercises are generally initiated with a low intensity and duration that increases progressively as healing progresses and tissue strength increases. It is also important to have established treatment goals or end points to guide progress.

The therapist may also develop preventive strategies for patients. These include monitoring the dog's weight, monitoring floor surfaces at home, having a preseason training period in athletic dogs, developing proper warm-up and cool down periods, recommendations for adequate rest between strenuous exercise sessions, and monitoring transportation between locations. For example, a dog with osteoarthritis of the stifle may benefit from weight loss, a judicious exercise plan with frequent exercises of short duration on soft surfaces, a padded surface to sleep, and a careful strategy of pharmacologic management, including analgesic medications and disease-modifying osteoarthritic agents.

Re-Evaluation and Assessment

The rehabilitation practitioner should regularly assess the dog's response to treatment, both during and after each session. During each session, the practitioner looks for adverse responses such as increased lameness or unwillingness to weight bear on a particular limb, fatigue, and increased anxiety. If any of these are apparent during a treatment session, the session may be altered or stopped. The overall treatment plan also may be modified if necessary. Many times, the dog will be given several days to rest and recover before continuing rehab. Following each session, the owners are instructed to watch for increased fatigue or abnormal pain behavior at home and to notify the practitioner and/or referring veterinarian if it does not resolve within 24 hours. In addition to observing response to rehabilitation, objective measurements and functional